Annotation-Informed Prior Scales for STBLR and MTBLR

Using posterior marker summaries and annotations to construct prior inclusion probabilities, prior effect variances, and sharing-model priors

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1 Motivation

In sparse Bayesian regression models fitted to genome-wide summary statistics, single-trait Bayesian linear regression (STBLR) can be used as a stable first-stage model. The resulting posterior marker summaries, especially posterior mean marker effects and posterior inclusion probabilities, can then be used to inform a second-stage multi-trait Bayesian linear regression (MTBLR) model.

The main idea is to use first-stage posterior summaries together with marker annotations or marker sets to construct annotation-informed prior parameters. These parameters can then be used in later STBLR or MTBLR analyses.

Typical first-stage outputs are:

$$\hat{b}_{it} = E(b_{it} \mid \text{data}),$$

and

$$\widehat{dm}_{it} = P(d_{it} = 1 \mid \text{data}),$$

where marker $i = 1, \dots, m$ and trait $t = 1, \dots, T$.

Let A denote an annotation matrix:

$$A_{ik} = \begin{cases} 1, & \text{if marker } i \text{ belongs to annotation } k, \\ 0, & \text{otherwise.} \end{cases}$$

The annotation columns may overlap. For example, a marker may be coding, close to a gene, and located inside a QTL region at the same time.

The goal is to derive annotation-specific or marker-specific prior parameters such as:

$$\pi_{k,t}, \qquad \sigma^2_{b,k,t}, \qquad B_k, \qquad \pi_{k,m},$$

where k indexes an annotation class or marker set, t indexes a trait, and m indexes a multi-trait sharing model.

2 Summary-statistic residual notation

Assume the STBLR or MTBLR model stores the following summary-statistic quantities:

$$wy_t = X^\top y_t,$$

$$ww_i = x_i^\top x_i,$$

and the residual score vector

$$r_t = X^\top y_t - X^\top X b_t.$$

Then

$$X^\top X b_t = wy_t - r_t.$$

This identity is useful because it allows approximate genetic variance and covariance decomposition using only summary-statistic quantities and marker effects.

For a single trait, the genomic variance estimate is

$$\widehat{V}_{G,t} = \frac{1}{n} b_t^\top X^\top X b_t = \frac{1}{n} b_t^\top (wy_t - r_t).$$

For traits t_1 and t_2, a symmetric genetic covariance estimate is

$$\widehat{G}_{t_1,t_2} = \frac{1}{2n}\left\{b_{t_1}^{\top}(wy_{t_2} - r_{t_2}) + b_{t_2}^{\top}(wy_{t_1} - r_{t_1})\right\}.$$

3 Set-specific genomic variance decomposition

For a marker set or annotation S_k, define the set-specific genomic variance contribution for trait t as

$$\widehat{V}_{G,k,t} = \frac{1}{n}\sum_{i\in S_k} \hat{b}_{it}(wy_{it} - r_{it}).$$

This gives a direct summary-statistic analogue of decomposing

$$b_t^{\top}X^{\top}Xb_t.$$

If the sets are non-overlapping, the set-specific contributions can be interpreted as an approximate partition of genomic variance. If the sets overlap, the contributions should be interpreted as annotation-specific attributions rather than a strict variance partition, because the same marker may contribute to more than one annotation.

3.1 R implementation

```
estimate_vg_by_set <- function(fit, sets, n) {
  b <- fit$bm
  q <- fit$wy - fit$r

  out <- matrix(NA_real_, nrow = length(sets), ncol = ncol(b))
  rownames(out) <- names(sets)
  colnames(out) <- colnames(b)

  for (s in seq_along(sets)) {
    idx <- sets[[s]]

    for (t in seq_len(ncol(b))) {
      out[s, t] <- sum(b[idx, t] * q[idx, t]) / n
    }
  }
```

```
  out
}
```

A proportion of total genomic variance can be computed as

```
vg_set <- estimate_vg_by_set(fitST, sets, n)
prop_of_vg <- sweep(vg_set, 2, colMeans(fitST$vgs), "/")
```

If phenotypes are standardized to variance approximately one, the values themselves can also be interpreted as approximate proportions of phenotypic variance.

4 Inclusion-probability enrichment

Annotation enrichment can also be assessed using posterior inclusion probabilities.

For annotation S_k, the expected number of included markers for trait t is

$$E[N_{k,t} \mid \text{data}] = \sum_{i \in S_k} \widehat{dm}_{it}.$$

A simple enrichment ratio is

$$\text{enrichment}_{k,t} = \frac{|S_k|^{-1} \sum_{i \in S_k} \widehat{dm}_{it}}{|S_k^c|^{-1} \sum_{i \notin S_k} \widehat{dm}_{it}}.$$

4.1 R implementation

```
estimate_pip_enrichment <- function(dm, sets) {
  out <- matrix(NA_real_, nrow = length(sets), ncol = ncol(dm))
  rownames(out) <- names(sets)
  colnames(out) <- colnames(dm)

  all_idx <- seq_len(nrow(dm))

  for (s in seq_along(sets)) {
    idx <- sets[[s]]
    bg <- setdiff(all_idx, idx)
```

```r
    for (t in seq_len(ncol(dm))) {
      mean_in <- mean(dm[idx, t])
      mean_bg <- mean(dm[bg, t])
      out[s, t] <- mean_in / mean_bg
    }
  }

  out
}
```

This is not a variance decomposition. It measures whether posterior inclusion probability is enriched in a marker set.

5 Combining effect size and inclusion probability

A useful LD-aware marker-level contribution is

$$c_{it} = \frac{\hat{b}_{it}(wy_{it} - r_{it})}{n}.$$

A posterior-inclusion-weighted contribution is

$$c_{it}^{\text{weighted}} = \frac{\widehat{dm}_{it}\hat{b}_{it}(wy_{it} - r_{it})}{n}.$$

This downweights markers with low posterior inclusion probability.

5.1 R implementation

```r
estimate_weighted_vg_by_set <- function(fit, sets, n) {
  b <- fit$bm
  dm <- fit$dm
  q <- fit$wy - fit$r

  out <- matrix(NA_real_, nrow = length(sets), ncol = ncol(b))
  rownames(out) <- names(sets)
  colnames(out) <- colnames(b)
```

```
  for (s in seq_along(sets)) {
    idx <- sets[[s]]

    for (t in seq_len(ncol(b))) {
      out[s, t] <- sum(dm[idx, t] * b[idx, t] * q[idx, t]) / n
    }
  }

  out
}
```

6 Handling overlapping annotations

When annotations overlap, direct set summation can double-count markers. A multiple-regression approach can be used to estimate partial annotation effects.

For a marker-level response s_i, fit

$$s_i = \alpha + \sum_k A_{ik}\gamma_k + e_i.$$

This estimates the effect of annotation k conditional on the other annotations.

Possible responses include:

$$s_{it} = \widehat{dm}_{it},$$

$$s_{it} = \hat{b}_{it}^2,$$

or

$$s_{it} = \frac{\hat{b}_{it}(wy_{it} - r_{it})}{n}.$$

For inclusion probabilities, a logit transformation can be used:

$$s_{it} = \text{logit}(\widehat{dm}_{it}).$$

6.1 R implementation: linear model on variance contribution

```
annotation_lm_vg <- function(fit, annot, trait, n) {
  b <- fit$bm[, trait]
  q <- fit$wy[, trait] - fit$r[, trait]

  y <- b * q / n
  X <- as.data.frame(annot)

  lm(y ~ ., data = X)
}
```

6.2 R implementation: weighted linear model

```
annotation_lm_vg_weighted <- function(fit, annot, trait, n, eps = 1e-6) {
  b <- fit$bm[, trait]
  dm <- fit$dm[, trait]
  q <- fit$wy[, trait] - fit$r[, trait]

  y <- b * q / n
  w <- pmax(dm, eps)
  X <- as.data.frame(annot)

  lm(y ~ ., data = X, weights = w)
}
```

6.3 R implementation: fractional response model for PIP

```
annotation_glm_dm <- function(dm, annot, trait, eps = 1e-6) {
  p <- pmin(pmax(dm[, trait], eps), 1 - eps)
  X <- as.data.frame(annot)

  glm(p ~ ., data = X, family = quasibinomial())
}
```

7 Using STBLR or MTBLR instead of ridge regression

Instead of a ridge regression on marker-level scores, one can fit STBLR or MTBLR with annotations as predictors.

Let S be a marker-by-trait matrix of marker-level scores. Examples are

$$S_{it} = \mathrm{logit}(\widehat{dm}_{it}),$$

$$S_{it} = \hat{b}_{it}^2,$$

or

$$S_{it} = \frac{\hat{b}_{it}(wy_{it} - r_{it})}{n}.$$

Then fit an annotation-level model:

$$S = A\Gamma + E,$$

where A is the marker-by-annotation design matrix and Γ is an annotation-by-trait coefficient matrix.

A Bayesian variable-selection prior on Γ can identify which annotations contribute to which traits. MTBLR can additionally learn whether annotation effects are shared across traits.

This turns overlapping-annotation adjustment into a Bayesian variable-selection problem.

8 Deriving annotation-specific prior inclusion probabilities

For annotation S_k, a simple empirical Bayes estimate of the inclusion probability for trait t is

$$\hat{\pi}_{k,t} = \frac{\alpha + \sum_{i \in S_k} \widehat{dm}_{it}}{2\alpha + |S_k|}.$$

Here α is a pseudo-count. A common choice is $\alpha = 1$.

8.1 R implementation

```
estimate_annotation_pi <- function(dm, sets, alpha = 1) {
  nt <- ncol(dm)
  K <- length(sets)

  pi_ann <- matrix(NA_real_, K, nt)
  rownames(pi_ann) <- names(sets)
  colnames(pi_ann) <- colnames(dm)

  for (k in seq_along(sets)) {
    idx <- sets[[k]]

    for (t in seq_len(nt)) {
      pi_ann[k, t] <- (alpha + sum(dm[idx, t])) / (2 * alpha + length(idx))
    }
  }

  pmin(pmax(pi_ann, 1e-8), 1 - 1e-8)
}
```

For marker-specific priors, assign each marker to an annotation or compute a marker-specific value from overlapping annotations.

9 Deriving annotation-specific prior effect variances

For annotation S_k, an estimated genomic variance contribution is

$$\widehat{V}_{G,k,t} = \frac{1}{n} \sum_{i \in S_k} \hat{b}_{it}(wy_{it} - r_{it}).$$

A simple prior scale is

$$\hat{\sigma}^2_{b,k,t} = \frac{\widehat{V}_{G,k,t}}{|S_k|}.$$

However, in sparse models it is usually better to divide by an effective number of active markers:

$$m^{\text{eff}}_{k,t} = \sum_{i \in S_k} \widehat{dm}_{it}.$$

Then

$$\hat{\sigma}^2_{b,k,t} = \frac{\widehat{V}_{G,k,t}}{\max(m^{\text{eff}}_{k,t}, c)},$$

where c is a floor such as 1 or 5.

9.1 R implementation

```
estimate_annotation_vb <- function(fit, sets, n, min_eff = 1, shrink = 0.5) {
  b  <- fit$bm
  dm <- fit$dm
  q  <- fit$wy - fit$r

  nt <- ncol(b)
  K <- length(sets)

  vb_ann <- matrix(NA_real_, K, nt)
  vg_ann <- matrix(NA_real_, K, nt)
  meff   <- matrix(NA_real_, K, nt)

  rownames(vb_ann) <- rownames(vg_ann) <- rownames(meff) <- names(sets)
  colnames(vb_ann) <- colnames(vg_ann) <- colnames(meff) <- colnames(b)

  for (k in seq_along(sets)) {
    idx <- sets[[k]]

    for (t in seq_len(nt)) {
      vg <- sum(b[idx, t] * q[idx, t]) / n
      neff <- sum(dm[idx, t])

      vg_ann[k, t] <- max(vg, 0)
      meff[k, t] <- neff
      vb_ann[k, t] <- max(vg_ann[k, t] / max(neff, min_eff), 1e-12)
    }
  }
```

```r
  # Shrink annotation-specific scales toward a global scale.
  vb_global <- colMeans(vb_ann, na.rm = TRUE)

  for (t in seq_len(nt)) {
    vb_ann[, t] <- shrink * vb_ann[, t] + (1 - shrink) * vb_global[t]
  }

  list(vb = vb_ann, vg = vg_ann, meff = meff)
}
```

10 Marker-specific prior scales for overlapping annotations

For overlapping annotations, marker-specific prior scales can be constructed on the log scale:

$$\log v_{b,i,t} = \mu_t + \sum_k A_{ik}\gamma_{k,t}.$$

Equivalently,

$$v_{b,i,t} = \exp\left(\mu_t + A_i\gamma_t\right).$$

The coefficients $\gamma_{k,t}$ may be estimated from an annotation regression or annotation-level BLR model.

Because annotation effects can be noisy, the multiplicative effect on the prior scale should be capped:

$$v_{b,i,t} = v_{b,t}^{\text{global}} \times \text{clip}\left(\exp(A_i\gamma_t), c_{\min}, c_{\max}\right).$$

10.1 R implementation

```r
derive_marker_vb_from_annotation_model <- function(global_vb, annot, gamma,
                                                   cap = c(0.25, 4)) {
  # annot: m x K annotation matrix
  # gamma: K x nt annotation coefficients on log scale
  # global_vb: length nt

  eta <- annot %*% gamma
```

```r
  vb_marker <- matrix(NA_real_, nrow(annot), ncol(gamma))

  for (t in seq_len(ncol(gamma))) {
    mult <- exp(eta[, t])
    mult <- pmin(pmax(mult, cap[1]), cap[2])
    vb_marker[, t] <- global_vb[t] * mult
  }

  colnames(vb_marker) <- colnames(gamma)
  vb_marker
}
```

11 Using marker-specific prior variances in STBLR

The standard BayesC marker update uses a global prior variance v_b. For marker i, the marginal Bayes factor is

$$\log BF_i = \frac{1}{2}\log\left(\frac{v_e}{v_e + w_i v_b}\right) + \frac{1}{2}\frac{s_i^2 v_b}{v_e(v_e + w_i v_b)},$$

where

$$s_i = r_i + w_i b_i.$$

With annotation-informed marker-specific prior variance $v_{b,i}$, replace v_b by $v_{b,i}$:

$$\log BF_i = \frac{1}{2}\log\left(\frac{v_e}{v_e + w_i v_{b,i}}\right) + \frac{1}{2}\frac{s_i^2 v_{b,i}}{v_e(v_e + w_i v_{b,i})}.$$

The posterior conditional distribution for b_i, given inclusion, becomes

$$b_i \mid d_i = 1, \cdots \sim N\left(\frac{s_i}{w_i + v_e/v_{b,i}}, \frac{v_e}{w_i + v_e/v_{b,i}}\right).$$

The corresponding C++ change is conceptually simple. Replace

```cpp
const double denom = ve + wi * vb;
const double lhs = wi + ve / vb;
```

with

```
const double vbi = vb_marker(i);
const double denom = ve + wi * vbi;
const double lhs = wi + ve / vbi;
```

12 Annotation-specific MTBLR prior covariance matrices

For MTBLR, marker effects are vectors:

$$\mathbf{b}_i = (b_{i1}, \dots, b_{iT})^\top.$$

The global MTBLR prior may use

$$\mathbf{b}_i \sim N(0, B).$$

An annotation-informed extension uses annotation-specific covariance matrices:

$$\mathbf{b}_i \mid a(i) = k \sim N(0, B_k).$$

A simple estimate of B_k is

$$\widehat{B}_k = \frac{\sum_{i \in S_k} w_i \hat{\mathbf{b}}_i \hat{\mathbf{b}}_i^\top}{\sum_{i \in S_k} w_i},$$

where w_i can be derived from posterior inclusion probabilities, for example

$$w_i = \max_t \widehat{dm}_{it}.$$

12.1 R implementation

```r
estimate_annotation_B <- function(bm, dm, sets, ridge = 1e-8, shrink = 0.5) {
  nt <- ncol(bm)
  K <- length(sets)

  B_list <- vector("list", K)
  names(B_list) <- names(sets)

  # Global covariance from posterior mean effects.
  w_global <- pmax(apply(dm, 1, max), 1e-6)
  B_global <- crossprod(bm * sqrt(w_global)) / sum(w_global)
  B_global <- B_global + diag(ridge, nt)

  for (k in seq_along(sets)) {
    idx <- sets[[k]]
    w <- pmax(apply(dm[idx, , drop = FALSE], 1, max), 1e-6)

    Bk <- crossprod(bm[idx, , drop = FALSE] * sqrt(w)) / sum(w)
    Bk <- shrink * Bk + (1 - shrink) * B_global
    Bk <- 0.5 * (Bk + t(Bk))
    diag(Bk) <- pmax(diag(Bk), ridge)

    B_list[[k]] <- Bk
  }

  B_list
}
```

13 Annotation-specific MTBLR model priors

In MTBLR, each marker can belong to one of several sharing models. For three traits, the model set may be

$$000, 100, 010, 001, 110, 101, 011, 111.$$

Let m_k be the binary vector for model k. A soft model responsibility based on STBLR inclusion probabilities is

$$w_{i,k} = \prod_{t=1}^{T} \widehat{dm}_{it}^{\,m_{k,t}} (1 - \widehat{dm}_{it})^{1-m_{k,t}}.$$

For annotation S_a, an annotation-specific sharing-model prior is

$$\pi_{a,k} = \frac{\alpha_k + \sum_{i \in S_a} w_{i,k}}{\sum_l \alpha_l + |S_a|}.$$

13.1 R implementation

```
estimate_annotation_model_pi <- function(dm, models, sets, alpha = 1) {
  M <- nrow(dm)
  nt <- ncol(dm)
  K <- length(sets)
  nmodels <- length(models)

  W <- matrix(0, M, nmodels)

  for (j in seq_len(nmodels)) {
    model <- models[[j]]
    p <- rep(1, M)

    for (t in seq_len(nt)) {
      dmt <- pmin(pmax(dm[, t], 1e-6), 1 - 1e-6)
      p <- p * if (model[t] == 1) dmt else (1 - dmt)
    }

    W[, j] <- p
  }

  pi_ann <- matrix(NA_real_, K, nmodels)
  rownames(pi_ann) <- names(sets)
  colnames(pi_ann) <- sapply(models, paste, collapse = "_")

  for (k in seq_along(sets)) {
    idx <- sets[[k]]
    counts <- colSums(W[idx, , drop = FALSE]) + alpha
    pi_ann[k, ] <- counts / sum(counts)
  }

  pi_ann
}
```

In the MTBLR marker update, a global model prior

```
loglik[k] = std::log(pi[k]) - 0.5 * logdet + 0.5 * quad;
```

can be replaced by an annotation-specific model prior:

```
loglik[k] = std::log(pi_annotation[ann_i][k]) - 0.5 * logdet + 0.5 * quad;
```

or a marker-specific prior:

```
loglik[k] = std::log(pi_marker(i, k)) - 0.5 * logdet + 0.5 * quad;
```

14 Prior scale derivation from annotation BLR

If an annotation-level BLR is fitted to marker-level scores, the posterior annotation effects can be used to construct prior scales.

For single-trait marker-specific effect variances:

$$\log v_{b,i,t} = \mu_t + A_i \gamma_t.$$

For marker-specific inclusion probabilities:

$$\operatorname{logit}(\pi_{i,t}) = \eta_t + A_i \alpha_t.$$

For MTBLR sharing-model priors:

$$\log \pi_{i,k} = \zeta_k + A_i \delta_k,$$

followed by normalization across models:

$$\pi_{i,k} = \frac{\exp(\zeta_k + A_i \delta_k)}{\sum_l \exp(\zeta_l + A_i \delta_l)}.$$

This provides a coherent way to use overlapping annotations without assigning each marker to a single class.

15 Recommended modelling hierarchy

A practical implementation can proceed in stages.

15.1 Stage 1: STBLR genome-wide screening

Run STBLR for each trait to obtain:

$$\hat{b}_{it}, \qquad \widehat{dm}_{it}, \qquad r_{it}.$$

These are used to estimate annotation effects, variance contributions, and sharing patterns.

15.2 Stage 2: Annotation summary and decomposition

Compute:

$$\widehat{V}_{G,k,t},$$

PIP enrichment,

$$\text{enrichment}_{k,t},$$

and, if desired, fit annotation-level STBLR or MTBLR models.

15.3 Stage 3: Derive prior parameters

Estimate:

$$\pi_{k,t},$$

$$\sigma^2_{b,k,t},$$

$$B_k,$$

and

$$\pi_{k,m}.$$

For overlapping annotations, derive marker-specific priors using log-linear or logistic annotation models.

15.4 Stage 4: Annotation-informed STBLR or MTBLR

Use the derived priors in a second-stage model:

$$b_{it} \sim (1 - \pi_{it})\delta_0 + \pi_{it} N(0, v_{b,it})$$

for STBLR, or

$$\mathbf{b}_i \sim \sum_m \pi_{i,m} N_m(0, B_i)$$

for MTBLR.

16 Warm-started regional MTBLR

The same STBLR outputs can also be used to warm-start MTBLR in selected regions.

Let

$$b_{it}^{(0)} = \hat{b}_{it}^{\text{STBLR}}.$$

Use this as the initial value for MTBLR.

Select candidate markers using STBLR posterior summaries, for example:

$$\max_t \widehat{dm}_{it} > \tau.$$

Then expand selected markers by LD neighborhoods and update only those markers in MTBLR.

The residual should still be initialized using all marker effects:

$$r_t = wy_t - X^\top X b_t^{(0)}.$$

Then, during MTBLR, update only markers inside selected regions while holding the remaining markers fixed at their STBLR posterior mean values.

This gives a computationally efficient approximate MTBLR analysis while preserving the genome-wide STBLR background effect.

17 Practical cautions

1. Annotation-derived prior parameters are empirical Bayes quantities. They should be documented as such.
2. Very small annotations can give unstable prior scale estimates. Use pseudo-counts, floors, and shrinkage toward global values.
3. For overlapping annotations, avoid interpreting direct set sums as a strict variance partition.
4. Marker-specific prior variances should be capped to avoid overly aggressive annotation effects.
5. If prior parameters are derived from the same data used for final inference, uncertainty may be underestimated. For prediction or discovery, this can still be useful, but for formal inference one may need cross-validation, sample splitting, or a fully hierarchical model.

18 Minimal example workflow

```
# 1. Run first-stage STBLR.
fitST <- stblr_cpg_omp_csr(...)

# 2. Estimate annotation-specific variance contributions.
vg_ann <- estimate_vg_by_set(fitST, sets, n)

# 3. Estimate annotation-specific inclusion probabilities.
pi_ann <- estimate_annotation_pi(fitST$dm, sets, alpha = 1)

# 4. Estimate annotation-specific prior effect variances.
vb_ann <- estimate_annotation_vb(fitST, sets, n, min_eff = 1, shrink = 0.5)

# 5. Estimate annotation-specific MTBLR sharing priors.
pi_model_ann <- estimate_annotation_model_pi(fitST$dm, models, sets, alpha = 1)

# 6. Optional: derive annotation-specific MT prior covariance matrices.
B_ann <- estimate_annotation_B(fitST$bm, fitST$dm, sets)

# 7. Use these quantities as priors, initial values, or region-selection weights
#    in a second-stage STBLR or MTBLR model.
```

19 Summary

STBLR posterior marker summaries can be used to construct biologically informed prior parameters for later STBLR or MTBLR analyses. The most useful derived quantities are:

$$\pi_{k,t},$$

annotation-specific inclusion probabilities,

$$\sigma^2_{b,k,t},$$

annotation-specific effect variances,

$$B_k,$$

annotation-specific multi-trait effect covariance matrices, and

$$\pi_{k,m},$$

annotation-specific multi-trait sharing-model probabilities.

For overlapping annotations, multiple regression, STBLR, or MTBLR can be used as an annotation-level model. This provides partial annotation effects and avoids arbitrary assignment of markers to a single category.

The resulting annotation-informed priors can be used for improved initialization, regional MTBLR fitting, marker-specific prior scales, and biologically interpretable multi-trait sharing analyses.

20 Empirical-Bayes priors from an annotation-level STBLR or MTBLR

A more coherent extension is to use the first genomic STBLR or MTBLR fit only as an intermediate step. The posterior marker summaries from this fit are converted into marker-level response variables. These response variables are then analysed with a second-stage STBLR or MTBLR model in which the design matrix is the marker-by-annotation matrix.

The complete empirical-Bayes loop is:

genomic STBLR/MTBLR → marker-level summaries → annotation STBLR/MTBLR → annotation-informed

This is different from using annotations directly in a ridge regression. The annotation-level STBLR or MTBLR can perform shrinkage, variable selection, and multi-trait sharing inference among annotations.

20.1 First-stage genomic fit

The first-stage genomic model is fitted using the ordinary summary-statistic inputs:

$$wy, \quad ww, \quad yy, \quad LD.$$

It produces marker-level posterior summaries:

$$\hat{b}_{it}, \qquad dm_{it}, \qquad r_{it}, \qquad q_{it} = wy_{it} - r_{it}.$$

Here $\hat{b}_{it}$ is the posterior mean marker effect, dm_{it} is the posterior inclusion probability, and q_{it} is the fitted marker score implied by the current posterior mean effects.

Useful marker-level response matrices include:

$$S_{it}^{(b)} = \hat{b}_{it},$$

$$S_{it}^{(pip)} = \text{logit}(dm_{it}),$$

and

$$S_{it}^{(vg)} = \frac{\hat{b}_{it}(wy_{it} - r_{it})}{n}.$$

The third response is LD-aware and directly related to the marker-level contribution to genomic variance.

In R:

```
S_b <- fit0$bm
S_pip <- qlogis(pmin(pmax(fit0$dm, 1e-6), 1 - 1e-6))
S_vg <- fit0$bm * (fit0$wy - fit0$r) / n
```

20.2 Annotation-level STBLR

Let A be the marker-by-annotation matrix. A single-trait annotation model is:

$$S_t = A\gamma_t + e_t,$$

where S_t is one column of a marker-level response matrix, and γ_t is a vector of annotation effects for trait t.

Using a BayesC-like prior,

$$\gamma_{kt} = \delta_{kt}\alpha_{kt},$$

$$\delta_{kt} \sim \text{Bernoulli}(\pi_{\gamma,t}),$$

and

$$\alpha_{kt} \mid \delta_{kt} = 1 \sim N(0, \sigma^2_{\gamma,t}).$$

The annotation-level STBLR posterior gives:

$$E(\gamma_{kt} \mid S),$$

and

$$P(\delta_{kt} = 1 \mid S).$$

These quantities measure whether annotation k explains marker-level evidence for trait t, conditional on other overlapping annotations.

20.3 Annotation-level MTBLR

For multiple traits, use:

$$S = A\Gamma + E,$$

where S is an $m \times T$ marker-level response matrix and Γ is a $K \times T$ matrix of annotation effects.

For annotation k:

$$\gamma_k = (\gamma_{k1}, \dots, \gamma_{kT})^\top.$$

An MTBLR prior can be used to infer annotation sharing patterns:

$$\gamma_k \sim \sum_m \pi_{\gamma,m} N_m(0, B_\gamma),$$

where m indexes trait-sharing models such as:

```
000, 100, 010, 001, 110, 101, 011, 111
```

for three traits. The posterior model probabilities indicate whether an annotation contributes to one trait, several traits, or all traits.

20.4 Deriving marker-specific inclusion priors

From an annotation model fitted to $S^{(pip)}$, construct a marker-specific annotation score:

$$\eta_{it} = \eta_{0t} + \sum_k A_{ik}\hat{\gamma}_{kt}.$$

Then transform it to a marker-specific prior inclusion probability:

$$\pi_{it} = \text{logit}^{-1}(\eta_{it}).$$

The annotation contribution should usually be centered so the genome-wide mean prior inclusion probability remains close to the original base rate.

```r
derive_pi_marker <- function(A, gamma, base_pi, cap = c(1e-8, 0.05)) {
  A <- as.matrix(A)
  gamma <- as.matrix(gamma)

  eta <- A %*% gamma

  # Center annotation contribution so the genome-wide mean stays near base_pi.
  eta <- sweep(eta, 2, colMeans(eta, na.rm = TRUE), "-")
  eta <- sweep(eta, 2, qlogis(base_pi), "+")

  pi_marker <- plogis(eta)
  pmin(pmax(pi_marker, cap[1]), cap[2])
}
```

A reasonable base rate is:

```r
base_pi <- colMeans(fit0$dm)
```

Then:

```r
pi_marker <- derive_pi_marker(
  A = annot,
  gamma = fitAnn_pip$bm,
  base_pi = base_pi,
  cap = c(1e-8, 0.05)
)
```

In the next genomic STBLR, marker i and trait t use π_{it} instead of a global π_t.

20.5 Deriving marker-specific prior effect variances

From an annotation model fitted to $S^{(vg)}$ or $S^{(b^2)}$, construct marker-specific prior effect variances:

$$\log v_{b,it} = \log v_{b,t}^{global} + \sum_k A_{ik}\hat{\theta}_{kt}.$$

Thus:

$$v_{b,it} = v_{b,t}^{global} \exp\left(\sum_k A_{ik}\hat{\theta}_{kt}\right).$$

Again, the annotation contribution should be centered and capped.

```
derive_vb_marker <- function(A, theta, global_vb, cap = c(0.25, 4)) {
  A <- as.matrix(A)
  theta <- as.matrix(theta)

  eta <- A %*% theta
  eta <- sweep(eta, 2, colMeans(eta, na.rm = TRUE), "-")

  mult <- exp(eta)
  mult <- pmin(pmax(mult, cap[1]), cap[2])

  vb_marker <- sweep(mult, 2, global_vb, "*")
  pmax(vb_marker, 1e-12)
}
```

For example:

```
global_vb <- colMeans(fit0$vbs)

vb_marker <- derive_vb_marker(
  A = annot,
  theta = fitAnn_vg$bm,
  global_vb = global_vb,
  cap = c(0.25, 4)
)
```

These marker-specific prior variances can be used in the next genomic STBLR by replacing the global v_b with $v_{b,it}$.

20.6 Deriving MTBLR marker-specific sharing-model priors

For MTBLR, annotation-level MTBLR can be used to derive marker-specific priors over sharing models.

Let η_{km} be the annotation-level effect of annotation k on sharing model m. Then for marker i:

$$\log \pi_{im} = \log \pi_m^{global} + \sum_k A_{ik}\eta_{km}.$$

Normalize across models:

$$\pi_{im} = \frac{\exp(\log \pi_{im})}{\sum_l \exp(\log \pi_{il})}.$$

```r
derive_model_pi_marker <- function(A, eta_model, base_pi, eps = 1e-12) {
  A <- as.matrix(A)
  eta_model <- as.matrix(eta_model)

  log_pi <- A %*% eta_model
  log_pi <- sweep(log_pi, 2, colMeans(log_pi, na.rm = TRUE), "-")
  log_pi <- sweep(log_pi, 2, log(pmax(base_pi, eps)), "+")

  log_pi <- sweep(log_pi, 1, apply(log_pi, 1, max), "-")
  pi_marker <- exp(log_pi)
  pi_marker <- sweep(pi_marker, 1, rowSums(pi_marker), "/")

  pmax(pi_marker, eps)
}
```

In the genomic MTBLR C++ marker sampler, this changes the model prior term from:

```cpp
loglik[k] = std::log(pi[k]) - 0.5 * logdet + 0.5 * quad;
```

to:

```cpp
loglik[k] = std::log(pi_marker(i, k)) - 0.5 * logdet + 0.5 * quad;
```

This allows annotations to influence whether a marker is more likely to be trait-specific or pleiotropic.

20.7 Deriving a global MTBLR prior covariance matrix from annotations

An annotation-level MTBLR also gives annotation effect estimates across traits. These can be used to estimate a global prior covariance matrix for the next genomic MTBLR run.

Let $\hat{\Gamma}$ be the matrix of annotation posterior mean effects. A weighted covariance estimate is:

$$B_{ann} = \frac{\sum_k w_k \hat{\gamma}_k \hat{\gamma}_k^\top}{\sum_k w_k},$$

where w_k can be an annotation-level posterior inclusion weight, for example:

$$w_k = \max_t dm_{kt}^{ann}.$$

```
estimate_B_from_annotation_model <- function(gamma, dm, ridge = 1e-8) {
  gamma <- as.matrix(gamma)
  w <- pmax(apply(dm, 1, max), 1e-6)

  B <- crossprod(gamma * sqrt(w)) / sum(w)
  B <- 0.5 * (B + t(B))
  diag(B) <- pmax(diag(B), ridge)

  B
}
```

This matrix can be used as the initial global B matrix in the next MTBLR run.

20.8 Recommended empirical-Bayes workflow

```
# ------------------------------------------------------------
# Round 1: genomic STBLR or MTBLR
# ------------------------------------------------------------
fit0 <- stblr_cpg_omp_csr(...)
# or:
# fit0 <- mtblr_cpg_omp_csr(...)

# Marker-level responses
S_b   <- fit0$bm
S_pip <- qlogis(pmin(pmax(fit0$dm, 1e-6), 1 - 1e-6))
S_vg  <- fit0$bm * (fit0$wy - fit0$r) / n
```

```r
# ------------------------------------------------------------
# Round 2: annotation-level STBLR or MTBLR
# ------------------------------------------------------------
# Here annot is the marker-by-annotation design matrix.
fitAnn_pip <- stblr_annotation(
  y = S_pip,
  X = annot
)

fitAnn_vg <- stblr_annotation(
  y = S_vg,
  X = annot
)

# Optional multi-trait annotation model.
fitAnn_mt <- mtblr_annotation(
  Y = S_vg,
  X = annot,
  models = models
)

# ------------------------------------------------------------
# Derive annotation-informed genomic priors
# ------------------------------------------------------------
pi_marker <- derive_pi_marker(
  A = annot,
  gamma = fitAnn_pip$bm,
  base_pi = colMeans(fit0$dm),
  cap = c(1e-8, 0.05)
)

vb_marker <- derive_vb_marker(
  A = annot,
  theta = fitAnn_vg$bm,
  global_vb = colMeans(fit0$vbs),
  cap = c(0.25, 4)
)

B_init <- estimate_B_from_annotation_model(
  gamma = fitAnn_mt$bm,
  dm = fitAnn_mt$dm
)
```

```r
# ------------------------------------------------------------
# Round 3: genomic STBLR or MTBLR with annotation-informed priors
# ------------------------------------------------------------
fit1 <- stblr_cpg_omp_csr(
  ...,
  pi_marker = pi_marker,
  vb_marker = vb_marker
)

fit1_mt <- mtblr_cpg_omp_csr(
  ...,
  B = B_init,
  pi_marker_model = pi_marker_model,
  adjE = 0.9
)
```

20.9 Interpretation

The annotation-level BLR or MTBLR does not estimate marker effects directly. It estimates how annotations explain marker-level evidence from the first genomic model. The resulting priors are therefore empirical-Bayes priors. They are useful for stabilizing later genomic analyses, improving initialization, guiding sharing-model probabilities, and incorporating biological information.

The most useful derived quantities are:

$$\pi_{it},$$

marker-specific inclusion probabilities,

$$v_{b,it},$$

marker-specific prior effect variances,

$$\pi_{im},$$

marker-specific MTBLR sharing-model probabilities, and

$$B_{ann},$$

an annotation-informed global MTBLR prior covariance matrix.

20.10 Cross-fitting and overfitting

Because the same data may be used to estimate marker summaries and then to construct priors, the procedure is empirical Bayes and can be optimistic if used for formal inference. A more conservative strategy is chromosome-wise cross-fitting:

```
for (chr in chromosomes) {
  # 1. Fit or summarize annotation model using all chromosomes except chr.
  # 2. Derive priors for markers on chr.
  # 3. Refit genomic STBLR or MTBLR on chr using those priors.
}
```

This avoids using the same markers both to estimate their annotation-informed priors and to evaluate their posterior evidence.

21 Direct annotation modelling inside the genomic STBLR or MTBLR sampler

The empirical-Bayes strategy above treats annotation modelling as a separate second-stage analysis. A more direct alternative is to include the annotation matrix inside the genomic STBLR or MTBLR sampler itself. In this formulation, annotations do not only summarize results after fitting; they directly parameterize the marker-level prior probabilities and prior scales used during sampling.

Let A be the marker-by-annotation matrix, with entries A_{ik} for marker $i = 1, \dots, m$ and annotation $k = 1, \dots, K$. The goal is to let A_i, the annotation profile of marker i, determine quantities such as:

$$\pi_{it},$$

$$v_{b,it},$$

$$\pi_{im},$$

and, for MTBLR,

$$B_i.$$

Here π_{it} is the prior inclusion probability for marker i and trait t, $v_{b,it}$ is the marker-specific prior effect variance, π_{im} is the marker-specific prior probability of MTBLR sharing model m, and B_i is a marker-specific prior covariance matrix for multivariate marker effects.

21.1 Direct annotation-informed STBLR

In a single-trait BayesC-style STBLR model, marker effects can be written as:

$$b_{it} = d_{it}\alpha_{it},$$

where

$$d_{it} \sim \text{Bernoulli}(\pi_{it}),$$

and

$$\alpha_{it} \mid d_{it} = 1 \sim N(0, v_{b,it}).$$

The annotation matrix can be used to model the prior inclusion probability as:

$$\text{logit}(\pi_{it}) = \mu_{\pi,t} + A_i\eta_{\pi,t},$$

where $\eta_{\pi,t}$ is a vector of annotation effects on inclusion probability for trait t.

Similarly, the marker-specific prior effect variance can be modelled as:

$$\log(v_{b,it}) = \mu_{v,t} + A_i\eta_{v,t}.$$

Equivalently,

$$v_{b,it} = \exp(\mu_{v,t} + A_i\eta_{v,t}).$$

This gives an annotation-aware STBLR model:

$$b_{it} \sim (1 - \pi_{it})\delta_0 + \pi_{it}N(0, v_{b,it}).$$

This is a direct extension of the usual global-prior STBLR model, where all markers share the same π_t and $v_{b,t}$.

21.2 Modified STBLR marker update

For marker i, define the partial residual score:

$$s_{it} = r_{it} + w_{it} b_{it},$$

where $w_{it} = x_i^\top x_i$ for trait t.

With marker-specific prior variance $v_{b,it}$ and residual variance $v_{e,t}$, the Bayes factor for inclusion becomes:

$$\log BF_{it} = \frac{1}{2} \log \left(\frac{v_{e,t}}{v_{e,t} + w_{it} v_{b,it}} \right) + \frac{1}{2} \frac{s_{it}^2 v_{b,it}}{v_{e,t}(v_{e,t} + w_{it} v_{b,it})}.$$

The posterior inclusion probability is:

$$P(d_{it} = 1 \mid -) = \frac{\pi_{it} BF_{it}}{(1 - \pi_{it}) + \pi_{it} BF_{it}}.$$

Conditional on inclusion, the posterior draw is:

$$b_{it} \mid d_{it} = 1, - \sim N(m_{it}, s_{it}^2),$$

where

$$m_{it} = \frac{s_{it}}{w_{it} + v_{e,t}/v_{b,it}},$$

and

$$s_{it}^2 = \frac{v_{e,t}}{w_{it} + v_{e,t}/v_{b,it}}.$$

Thus, the only change in the existing STBLR sampler is to replace the global π_t and $v_{b,t}$ by marker-specific values π_{it} and $v_{b,it}$.

A corresponding C++ change is conceptually:

```
const double pi1 = pi_marker(i);
const double pi0 = 1.0 - pi1;
const double vbi = vb_marker(i);

const double denom = ve + wi * vbi;

const double logBF =
  0.5 * std::log(ve / denom) +
  0.5 * score * score * vbi / (ve * denom);

const double lhs = wi + ve / vbi;
const double mean = score / lhs;
const double sd = std::sqrt(ve / lhs);
```

For multiple traits running in parallel STBLR, π_{it} and $v_{b,it}$ can be stored as marker-by-trait matrices:

```
const double pi1 = pi_marker(i, t);
const double vbi = vb_marker(i, t);
```

21.3 Estimating annotation effects inside the sampler

There are two main ways to handle the annotation coefficients $\eta_{\pi,t}$ and $\eta_{v,t}$.

21.3.1 Fixed annotation effects

The simplest approach is to estimate $\eta_{\pi,t}$ and $\eta_{v,t}$ before the genomic sampler, for example from a previous STBLR/MTBLR annotation analysis, and then treat them as fixed during the next genomic run.

This gives a stable empirical-Bayes model:

$$\hat{\eta}_{\pi,t}, \hat{\eta}_{v,t} \quad \Rightarrow \quad \pi_{it}, v_{b,it} \quad \Rightarrow \quad \text{annotation-informed genomic STBLR.}$$

This is the easiest and safest implementation.

21.3.2 Updating annotation effects during sampling

A more ambitious approach is to update the annotation effects inside the sampler. For example, one could place priors on annotation effects:

$$\eta_{\pi,kt} \sim N(0, \sigma^2_{\eta_\pi,t}),$$

and

$$\eta_{v,kt} \sim N(0, \sigma^2_{\eta_v,t}).$$

The sampler would then alternate between:

1. marker-effect updates using current π_{it} and $v_{b,it}$,
2. annotation-effect updates using current marker inclusion states d_{it} and marker effects b_{it},
3. variance component updates.

For inclusion probabilities, the annotation update can be viewed as a Bayesian logistic regression:

$$d_{it} \sim \text{Bernoulli}(\pi_{it}),$$

with

$$\text{logit}(\pi_{it}) = \mu_{\pi,t} + A_i \eta_{\pi,t}.$$

For effect variances, active marker effects can inform a log-variance regression:

$$\log(b^2_{it} + \epsilon) \approx \mu_{v,t} + A_i \eta_{v,t} + e_{it},$$

using only markers with $d_{it} = 1$, or using weights based on posterior inclusion probabilities.

This fully Bayesian version is more elegant, but also more complicated and more prone to mixing issues. For implementation, it is usually better to start with fixed annotation-derived priors.

21.4 Direct annotation-informed MTBLR

In MTBLR, each marker has a vector of trait effects:

$$\mathbf{b}_i = (b_{i1}, \dots, b_{iT})^\top.$$

The usual MTBLR model uses a set of sharing models M_m. For example, with three traits the model space may be:

$$000, 100, 010, 001, 110, 101, 011, 111.$$

A direct annotation-aware MTBLR model lets the annotation profile A_i determine the prior probability of each sharing model:

$$P(M_i = m \mid A_i) = \pi_{im}.$$

A convenient parameterization is a multinomial-logit model:

$$\log \pi_{im} = c_m + A_i \eta_m - C_i,$$

where C_i is the normalizing constant:

$$C_i = \log \left(\sum_l \exp(c_l + A_i \eta_l) \right).$$

Equivalently,

$$\pi_{im} = \frac{\exp(c_m + A_i \eta_m)}{\sum_l \exp(c_l + A_i \eta_l)}.$$

The annotation matrix can therefore make coding markers more likely to follow a shared model, promoter markers more likely to be trait-specific, or QTL markers more likely to have non-null effects across several traits.

21.5 MTBLR model prior in the marker update

In the MTBLR marker sampler, the model posterior weight for marker i and model m typically has the form:

$$\log w_{im} = \log \pi_m + \log L_{im},$$

where $\log L_{im}$ is the marginal likelihood contribution under sharing model m.

With annotation-informed marker-specific model priors, this becomes:

$$\log w_{im} = \log \pi_{im} + \log L_{im}.$$

Thus, the existing sampler only needs to replace the global model prior π_m with marker-specific priors π_{im}.

In C++ this corresponds to replacing:

```
loglik[k] = std::log(pi[k]) - 0.5 * logdet + 0.5 * quad;
```

with:

```
loglik[k] = std::log(pi_marker_model(i, k)) - 0.5 * logdet + 0.5 * quad;
```

This is likely the most useful direct annotation extension for MTBLR, because it lets annotations guide the sharing model while still allowing the multivariate likelihood to override the prior when the data are strong.

21.6 Annotation-informed MTBLR effect covariance

Annotations can also affect the prior covariance matrix of marker effects. A global MTBLR model may use:

$$\mathbf{b}_i \mid M_i = m \sim N_m(0, B),$$

where $N_m(0, B)$ denotes a multivariate normal distribution restricted to the active traits in model m.

An annotation-aware version can use marker-specific or annotation-specific covariance matrices:

$$\mathbf{b}_i \mid M_i = m, A_i \sim N_m(0, B_i).$$

A log-linear construction for trait-specific prior variances is:

$$\log B_{i,tt} = \mu_{B,t} + A_i \eta_{B,t}.$$

For cross-trait covariances, one can either use a global correlation matrix R and annotation-specific standard deviations:

$$B_i = D_i R D_i,$$

where

$$D_i = \text{diag}(\sqrt{B_{i,11}}, \dots, \sqrt{B_{i,TT}}),$$

or use annotation-specific full covariance matrices B_k.

The first approach is more stable:

$$B_i = D_i R D_i.$$

It lets annotations change the amount of prior variance per trait while preserving a globally estimated cross-trait correlation structure.

The second approach is more flexible:

$$B_i = \sum_k w_{ik} B_k,$$

where w_{ik} are normalized annotation weights for marker i. However, this can be unstable if annotations are sparse or highly overlapping.

21.7 Direct annotation-aware STBLR and MTBLR as hierarchical models

The direct annotation model can be summarized as a hierarchical prior.

For STBLR:

$$
\begin{aligned}
\operatorname{logit}(\pi_{it}) &= \mu_{\pi,t} + A_i\eta_{\pi,t}, \\
\log(v_{b,it}) &= \mu_{v,t} + A_i\eta_{v,t}, \\
d_{it} &\sim \text{Bernoulli}(\pi_{it}), \\
b_{it} \mid d_{it} = 1 &\sim N(0, v_{b,it}), \\
b_{it} \mid d_{it} = 0 &= 0.
\end{aligned}
$$

For MTBLR:

$$
\begin{aligned}
\pi_{im} &= \frac{\exp(c_m + A_i\eta_m)}{\sum_l \exp(c_l + A_i\eta_l)}, \\
M_i &\sim \text{Categorical}(\pi_{i1}, \dots, \pi_{iM}), \\
\mathbf{b}_i \mid M_i = m, A_i &\sim N_m(0, B_i).
\end{aligned}
$$

These formulations are direct annotation-informed genomic models rather than post hoc annotation analyses.

21.8 Practical implementation strategy

A good implementation path is incremental.

21.8.1 Step 1: Fixed marker-specific priors

First, compute π_{it}, $v_{b,it}$, and π_{im} outside the sampler, then pass them as inputs:

```
fit <- stblr_cpg_omp_csr(
  ...,
  pi_marker = pi_marker,
  vb_marker = vb_marker
)

fit_mt <- mtblr_cpg_omp_csr(
  ...,
  pi_marker_model = pi_marker_model
)
```

This only requires small changes to the marker-update functions.

21.8.2 Step 2: Annotation matrix as input, fixed annotation coefficients

Instead of passing full marker-specific prior matrices, pass the annotation matrix and annotation coefficients:

```
fit <- stblr_cpg_omp_csr(
  ...,
  annot = A,
  eta_pi = eta_pi,
  eta_vb = eta_vb
)
```

Then compute marker-specific priors inside C++:

$$\pi_{it} = \text{logit}^{-1}(\mu_{\pi,t} + A_i \eta_{\pi,t}),$$

and

$$v_{b,it} = \exp(\mu_{v,t} + A_i \eta_{v,t}).$$

This saves memory if m is large and the annotation matrix is sparse.

21.8.3 Step 3: Update annotation coefficients

Finally, update $\eta_{\pi,t}$, $\eta_{v,t}$, or η_m inside the sampler. This creates a fully hierarchical annotation-informed BLR/MTBLR model, but requires additional MCMC steps, careful priors, and good diagnostics.

21.9 Sparse annotation matrix implementation

For large numbers of markers and annotations, the annotation matrix A should be stored sparsely. For each marker i, store the list of active annotations:

```cpp
struct AnnotationNeighbor {
  int k;
  float value;
};

using AnnotationRows = std::vector<std::vector<AnnotationNeighbor>>;
```

Then the linear predictor for marker i is:

```cpp
double eta_i = intercept;
for (const auto& a : annot_rows[i]) {
  eta_i += static_cast<double>(a.value) * eta[a.k];
}
```

For STBLR:

```cpp
const double pi_i = inv_logit(eta_pi_i);
const double vb_i = std::exp(eta_vb_i);
```

For MTBLR sharing models:

```cpp
for (int model = 0; model < nmodels; ++model) {
  log_pi_model[model] = base_log_pi[model];
  for (const auto& a : annot_rows[i]) {
    log_pi_model[model] += a.value * eta_model(a.k, model);
  }
}

// Then apply softmax to get pi_marker_model[i, model].
```

This avoids constructing dense marker-by-model or marker-by-trait prior matrices when m is large.

21.10 Centering, shrinkage, and caps

Direct annotation priors can be powerful but need regularization. The annotation contribution should usually be centered so that it redistributes prior mass rather than changing the total prior sparsity too aggressively.

For inclusion probabilities:

$$\eta_{it}^{centered} = A_i \eta_{\pi,t} - \frac{1}{m} \sum_{j=1}^{m} A_j \eta_{\pi,t}.$$

Then:

$$\text{logit}(\pi_{it}) = \text{logit}(\pi_t^{global}) + \eta_{it}^{centered}.$$

For prior variances:

$$\log(v_{b,it}) = \log(v_{b,t}^{global}) + \xi_{it}^{centered},$$

where

$$\xi_{it}^{centered} = A_i \eta_{v,t} - \frac{1}{m} \sum_{j=1}^{m} A_j \eta_{v,t}.$$

The resulting multipliers should be capped:

$$\frac{v_{b,it}}{v_{b,t}^{global}} \in [c_{min}, c_{max}],$$

for example:

$$[c_{min}, c_{max}] = [0.25, 4].$$

Similarly, marker-specific inclusion probabilities should be bounded:

$$\pi_{it} \in [\epsilon, \pi_{max}],$$

for example:

$$\epsilon = 10^{-8}, \qquad \pi_{max} = 0.05.$$

These constraints prevent a highly enriched annotation from overwhelming the likelihood or causing too many markers to enter the model.

21.11 Recommended first direct model

The most practical first direct annotation model is:

$$\operatorname{logit}(\pi_{it}) = \operatorname{logit}(\pi_t^{global}) + A_i \eta_{\pi,t}^{centered},$$

with a global $v_{b,t}$ unchanged.

For MTBLR, the most practical first model is:

$$\log \pi_{im} = \log \pi_m^{global} + A_i \eta_m^{centered} - C_i.$$

These two extensions are simple, interpretable, and likely to be stable. Marker-specific prior variances $v_{b,it}$ and marker-specific covariance matrices B_i are useful next steps, but they should be introduced with shrinkage and caps.

21.12 Causal interpretation and causal-hypothesis generation

The annotation-informed STBLR/MTBLR framework can provide evidence for genetic sharing, annotation enrichment, and plausible biological mechanisms, but it does not by itself identify causal relationships in the intervention sense. The model can estimate posterior quantities such as

$$P(d_{it} = 1 \mid \text{data}),$$

where d_{it} indicates that marker i is active for trait t, and in the multivariate case

$$P(M_i = m \mid A_i, \text{data}),$$

where M_i is a trait-sharing model and A_i denotes the annotation profile of marker i. These are statements about which genetic effects and annotation patterns best explain the observed summary statistics. They are not, by themselves, causal effects such as

$$\text{annotation} \to \text{trait},$$

or

$$\text{trait } 1 \to \text{trait } 2.$$

A careful interpretation is that the model identifies annotations, regions, and marker-sharing configurations that are enriched for genetic signal. For example, if annotation A has high posterior support for shared effects on traits t_1 and t_2, then the model supports the statement

$$A \text{ is enriched for shared genetic effects on } t_1 \text{ and } t_2.$$

This is a genetic-architecture statement, not a direct causal claim. It can, however, motivate causal hypotheses about biological mechanisms.

21.12.1 Genetic sharing versus causal direction

MTBLR can help distinguish different forms of genetic sharing. For example, a marker may support a sharing model such as

$$M_i = 110,$$

which indicates an effect on traits 1 and 2 but not trait 3. If this sharing pattern is enriched in annotation A, then

$$P(M_i = 110 \mid A_i, \text{data})$$

may be larger than the genome-wide background probability. This suggests that annotation A may tag a biological mechanism relevant to both traits.

However, this does not determine whether the relationship is

$$G \to t_1 \to t_2,$$

or

$$G \to t_2 \to t_1,$$

or horizontal pleiotropy,

$$G \to t_1 \quad \text{and} \quad G \to t_2,$$

through separate biological pathways. Additional information is required to infer directionality.

21.12.2 What can be inferred directly

The framework can directly support posterior statements such as:

$$P(\text{marker } i \text{ is active for trait } t \mid \text{data}),$$

$$P(\text{marker } i \text{ has a shared effect across traits} \mid \text{data}),$$

$$P(\text{annotation } k \text{ is enriched for effects} \mid \text{data}),$$

and

$$P(M_i = m \mid A_i, \text{data}).$$

These quantities can be used to prioritize markers, regions, annotations, and biological pathways for downstream causal analyses. They should be interpreted as posterior evidence for association and sharing, not as proof of causation.

21.12.3 Connection to colocalization

A natural downstream analysis is colocalization. Suppose a genomic region shows a strong MTBLR sharing signal between a molecular trait x and a complex trait y. If fine-mapping and colocalization support the same causal variant for both traits, then this strengthens the hypothesis

$$G \to x \to y.$$

In this setting, annotation-aware MTBLR can be used as a prioritization step. It can identify regions and annotations where shared effects are likely, and colocalization can then test whether the same underlying variant is responsible for the molecular and complex-trait associations.

21.12.4 Connection to Mendelian randomization

Mendelian randomization can be used to test directional hypotheses such as

$$x \to y,$$

using genetic variants as instruments. The validity of this interpretation depends on instrumental-variable assumptions. In particular, the genetic instruments should be associated with the exposure, should not be associated with confounders of the exposure-outcome relationship, and should affect the outcome only through the exposure.

Annotation-aware MTBLR can help select or classify candidate instruments. For example, variants that appear trait-specific for exposure x may be more useful as instruments than variants with strong evidence of broad horizontal pleiotropy. Conversely, markers assigned to shared-effect models may indicate biological pleiotropy and should be handled carefully in causal analyses.

21.12.5 Annotation-specific causal hypotheses

The most useful causal-hypothesis output from the framework may be annotation-specific sharing probabilities. For example, for annotation k, one can estimate

$$P(M = 110 \mid A_k = 1, \text{data}),$$

or compare it to the background probability

$$P(M = 110 \mid \text{data}).$$

If

$$P(M = 110 \mid A_k = 1, \text{data}) > P(M = 110 \mid \text{data}),$$

then annotation k is enriched for variants with shared effects on traits 1 and 2. This can motivate a biological hypothesis that the functional class represented by annotation k is involved in a mechanism linking the two traits.

Similarly, if annotation k is enriched for trait-specific models, such as

$$M = 100,$$

then the annotation may be more relevant to a trait-specific pathway than to a shared pathway.

21.12.6 Practical causal workflow

A practical workflow is:

1. Fit STBLR or MTBLR using GWAS summary statistics and sparse LD.
2. Estimate marker-level posterior summaries such as bm, dm, and sharing-model probabilities.
3. Use annotation-aware STBLR/MTBLR to identify annotations enriched for trait-specific or shared genetic effects.
4. Use the enriched annotations and regions to define candidate mechanisms.
5. Perform regional fine-mapping to construct credible sets.
6. Test colocalization with molecular QTL data, such as expression, methylation, protein, metabolite, or microbiome QTL.
7. Use Mendelian randomization, mediation analysis, perturbation data, or experimental validation to test directional causal hypotheses.

Thus, the annotation-informed STBLR/MTBLR framework is best viewed as a mechanism-discovery and hypothesis-prioritization layer. It can point to annotations and regions that are likely to be biologically relevant, but causal interpretation requires additional assumptions or external evidence.

A suitable manuscript statement is:

> The annotation-informed STBLR/MTBLR framework identifies annotations and genomic regions enriched for trait-specific or shared genetic effects. These results can prioritize plausible causal mechanisms and candidate mediators. However, causal interpretation requires additional assumptions or complementary analyses, such as fine-mapping, colocalization, Mendelian randomization, mediation analysis, perturbation experiments, or external functional validation.